1 Introduction

We saw in the previous section that

- If V is a finite-dimensional vector space, $T \in \mathscr{L}(V)$, and B, B' bases for V, then $[T]_{B'}$ and $[T]_B$ are similar matrices:

  $$[T]_{B'} = C[T]_B C^{-1},$$

  where C is the change of basis matrix from $B \to B'$.

- Similar matrices all have the same determinant, trace, rank, and nullity, which allow us to define these quantities for $T \in \mathscr{L}(V)$ as the corresponding quantity for $[T]_B$ in any basis.

Given $T \in \mathscr{L}(V)$, we therefore seek a basis in which $[T]_B$ takes a form which is as simple as possible. In particular, if we can find a basis in which $[T]_B$ is diagonal, then the determinant, trace, rank, and nullity can all be found immediately by inspection.

Exercise 1.1. Suppose a linear operator T can be represented in some basis by the matrix

$$[T]_B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Then

(a) $\det T =$

(b) $\operatorname{tr} T =$

(c) $\operatorname{rank} T =$

(d) $\dim \ker T =$

(e) Bases for each of the Fundamental Matrix Spaces are given by

The previous example showed that if we can find a basis in which T is represented by a diagonal matrix, we can readily answer any question about the properties of T. This leads to the following questions:

(1) Can each linear operator $T \in \mathscr{L}(V)$ be represented by a diagonal matrix in some basis?

(2) If not, for which operators T does such a basis exist?

(3) How can be find such a basis if there is one?

(4) If no such basis exists, what is the simplest type of matrix by which we can represent T?

We begin by noting the following.

Theorem 1.2 (Diagonality condition)**.** *Let V be a finite-dimensional vector space, $B = (v_1, v_2, \ldots, v_n)$ an ordered basis for V, and $T \in \mathscr{L}(V)$. Then*

$$[T]_B = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

is diagonal if and only if $T(v_i) = \lambda_i v_i$ for each $i = 1, \cdots, n$.

Exercise 1.3. Prove Theorem 1.2.

Definition 1.4 (Eigenvalue of a linear operator)**.** Let V be a vector space and $T \in \mathscr{L}(V)$. An *eigenvalue* of T is a scalar λ such that there is a non-zero vector $v \in V$ such that

$$T(v) = \lambda v.$$

The vector v is called an *eigenvector* of T associated with the eigenvalue λ.

Note that, if v is an eigenvector of T associated with eigenvalue λ and if B is a basis for V, then

$$[T(v)]_B = [T]_B[v]_B = [\lambda v]_B = \lambda [v]_B.$$

Definition 1.5 (Eigenvalue of an $n \times n$ matrix)**.** An eigenvalue of an $n \times n$ matrix A is a scalar λ such that there exists a nonzero vector $v \in \mathbb{R}^n$ such that

$$Av = \lambda v$$

The vector v is called an *eigenvector* of A associated with the eigenvalue λ.

Exercise 1.6. Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, $u = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, and $v = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$. Are u and v eigenvectors of A? If so, find the corresponding eigenvalues.

Proposition 1.7 (Similar matrices have the same eigenvalues)**.** *Similar matrices have the same eigenvalues.*

Exercise 1.8. Prove Proposition 1.7.

Corollary 1.9. *Let $T \in \mathscr{L}(V)$. Then any matrix representing T has the same eigenvalues as T.*

We now discuss how to systematically find eigenvalues of a given matrix. A scalar λ is an eigenvalue of A if and only if $Av = \lambda v$ has a nontrivial solution. This is equivalent to the homogeneous linear system

$$(A - \lambda I)v = 0, \tag{1.10}$$

where I is the $n \times n$ identity matrix.

Definition 1.11 (Eigenspace)**.** The set of all solutions of Equation (1.10) is called the *eigenspace* of A associated to the eigenvalue λ, which we will denote by E_λ.

Proposition 1.12. *Suppose $A \in M^{n\times n}(\mathbb{R})$ has an eigenvalue λ. Then the eigenspace of A corresponding to λ is a subspace of $\mathbb{R}^n$.*

Exercise 1.13. Prove Proposition 1.12.

Note that the equation $(A - \lambda I)v = \mathbf{0}$ has a nontrivial solution if and only if $\det(A - \lambda I) = 0$, or, equivalently, if $\det(\lambda I - A) = 0$.

Theorem 1.14. *Let A be an $n \times n$ matrix. Then* $\det(\lambda I - A)$ *is a monic (leading coefficient is 1), degree n polynomial.*

Proof. (By induction on n). Let A be a 2×2 matrix. Then

$$\begin{aligned}\det(\lambda I - A) = \begin{vmatrix} \lambda - a_{11} & -a_{12} \\ -a_{21} & \lambda - a_{22} \end{vmatrix} &= (\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21} \\ &= \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21}\end{aligned}$$

is a monic, degree 2 polynomial. Suppose the assertion is true for $n = k$, and let A be a $(k+1) \times (k+1)$ matrix. Then

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - a_{11} & a_{12} & \dots & a_{1,k+1} \\ a_{21} & \lambda - a_{22} & \dots & a_{a,k+1} \\ \vdots & & & \\ a_{k+1,1} & a_{k+2,2} & \dots & \lambda - a_{k+1,k+1} \end{vmatrix}.$$

Evaluating the determinant by cofactor expansion along the first row, we obtain

$$\det(\lambda I - A) = (\lambda - 1_{11})C_{11} + a_{12}C_{12} + \dots a_{k+1}C_{1,k+1}.$$

Since $C_{1j} = (-1)^{1+j} \det A_{1j}$ with A_{1j} $k \times k$, by the inductive hypothesis $C_{1j} = (-1)^{1+j}(\lambda^n + \dots)$, where the $\dots$ denote lower-degree terms in λ. Thus,

$$\det(\lambda I - A) = (\lambda - a_{11})(\lambda^k + \dots) + a_{12}(-\lambda^k + \dots) + \dots a_{k+1}(-1)^{k+1}(\lambda^k + \dots) = \lambda^{k+1} + \dots,$$

which proves the assertion holds for all $n \in \mathbb{N}$. □

Definition 1.15 (Characteristic equation)**.** Given any $n \times n$ matrix A, the monic, degree n polynomial $\det(\lambda I - A)$ is called the *characteristic polynomial* of A. The equation

$$\det(\lambda I - A) = 0$$

is called the *characteristic equation* of A.

Example 1.16. Let $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Then the characteristic equation for A is

$$0 = \det(\lambda I - A) = \begin{vmatrix} \lambda & 1 \\ -1 & \lambda \end{vmatrix} = \lambda^2 + 1,$$

which has solution $\pm i$.

Theorem 1.17. *If V is a complex vector space, then every linear operator has an eigenvalue (and thus an eigenvector).*

Proof. By the Fundamental Theorem of Algebra, the polynomial equation $\det(\lambda I - A) = 0$ has a root in $\mathbb{C}$ for any matrix A representing T. □

We see from Example 1.16 that this theorem is false if V is a real vector space.

Corollary 1.18. *An $n \times n$ matrix A has at most n distinct eigenvalues.*

Proof. By the Fundamental Theorem of Algebra, every non-constant polynomial has n roots in $\mathbb{C}$ (counted with multiplicity). □

The next example shows how to find the eigenvalues and corresponding eigenspaces of a given $n \times n$ matrix.

Example 1.19. Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$. The characteristic equation of A is given by

$$\begin{aligned} 0 &= \begin{vmatrix} 1-\lambda & 6 \\ 5 & 2-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda) - 30 \\ &= 2 - 3\lambda + \lambda^2 - 30 \\ &= \lambda^2 - 3\lambda - 28 \\ &= (\lambda - 7)(\lambda + 4) \end{aligned}$$

so A has two eigenvalues $\lambda = 7, -4$. To find a parametric description of W_7, we row reduce the matrix

$$A - 7I = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix}$$

giving

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix},$$

which shows that $E_7 = \text{span}\,\{(1,1)\}$. Geometrically, this is the line $y = x$ in the plane.

Exercise 1.20. Work out E_{-4} from the previous example.

Proposition 1.21. *Similar matrices have the same characteristic equation.*

Prove Propsition 1.21.

WARNING! We have seen that similar matrices have the same eigenvalues. However, the converse is not true: two matrices which have the same eigenvalues are not necessarily similar, as the next example shows.

Example 1.22. Consider the matrices $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$. We note that $B = 2I$. If C is any matrix similar to B, then there exists an invertible matrix P such that $C = P^{-1}BP$. But then $C = P^{-1}BP = P^{-1}2IP = 2P^{-1}IP = 2I = B$. Thus, the only matrix similar to B is B itself. In particular, A is not similar to B. However, A and B have the same characteristic polynomials and therefore the same eigenvalues.

Computing eigenvalues of larger matrices require solving polynomial equations of degree n. If a matrix A is triangluar, then this is easy.

Proposition 1.23. *The eigenvalues of a triangular matrix are the entries on the main diagonal.*

Proof. Let

$$A = \begin{bmatrix} a_{11} & * & * & \dots & * \\ 0 & a_{22} & * & \dots & * \\ 0 & 0 & \ddots & \dots & * \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a_{nn} \end{bmatrix}.$$

Then the characteristic equation

$$0 = \det(\lambda I - A) = \begin{vmatrix} \lambda - a_{11} & * & * & \dots & * \\ 0 & \lambda - a_{22} & * & \dots & * \\ 0 & 0 & \ddots & \dots & * \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \dots \lambda - & a_{nn} \end{vmatrix} = \prod_{i=1}^{n} (\lambda - a_{ii})$$

has solutions $a_{11}, a_{22}, \dots, a_{nn}$. □

Example 1.24. The distinct eigenvalues of

$$A = \begin{bmatrix} 5 & -2 & 6 & 1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

are 1,3,5.

Exercise 1.25. Show that the eigenspaces associated to the eigenvalues in the previous example are

$$W_1 = \text{span}\left\{\begin{bmatrix} -1/4 \\ -4 \\ -1 \\ 1 \end{bmatrix}\right\},\ W_3 = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}\right\},\ W_5 = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}\right\}.$$

Unfortunately, solving for the roots of an arbitrary polynomial is a hard problem, so in general eigenvalues are difficult to compute. However, computers can approximate the eigenvalues of a given matrix to very good precision.

Finding eigenvalues of 3×3 matrices or larger (which may not be triangular) with *integer* entries can sometimes be accomplished with the help of the following theorem:

Theorem 1.26 (Rational root theorem)**.** *Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a degree n polynomial with integer coefficients. If $x = \frac{p}{q}$ is a* rational *root of $p(x)$, then $p \mid a_0$ and $q \mid a_n$.* [1]

Proof. Assume $x = \frac{p}{q}$ is a root of $p(x) = 0$. Assume also that $\frac{p}{q}$ is in lowest terms, so that p and q have no common factors. Then

$$a_n \frac{p^n}{q^n} + a_{n-1}\frac{p^{n-1}}{q^{n-1}} + \cdots + a_1 \frac{p}{q} + a_0 = 0.$$

Multiplying both sides by q^n, we obtain

$$a_n p^n + a_{n-1}p^{n-1}q + a_{n-2}q^{n-2}q^2 + \cdots + a_1 pq^{n-1} + a_0 q^n = 0. \tag{1.27}$$

and therefore

$$a_n p^n + a_{n-1}p^{n-1}q + a_{n-2}q^{n-2}q^2 + \cdots + a_1 pq^{n-1} = -a_0 q^n.$$

The left hand side is divisible by p (since each term has at least one factor of p), so the right hand side must be as well. Since p does not divide q, it also does not divide q^n, and therefore p must divide a_0.

We can also rewrite Equation (1.27) as

$$a_{n-1}p^{n-1}q + a_{n-2}q^{n-2}q^2 + \cdots + a_1 pq^{n-1} + a_0 q^n = -a_n p^n + .$$

Since the left-hand side is divisible by q, the right-hand side must be as well. Since q does not divide p, it also does not divide p^n and therefore q must divide a_n, completing the proof. □

Note that Theorem 1.26 does *not* say that every polynomial with integer coefficients has a rational root. It only says that *if* such a polynomial has a rational root, then it must be of the form in the theorem.

Example 1.28. Let $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{bmatrix}$. The characteristic equation of A is given by

$$0 = \det(\lambda I - A) = \begin{vmatrix} 3-\lambda & 1 & -1 \\ 2 & 2-\lambda & -1 \\ 2 & 2 & -\lambda \end{vmatrix} = \lambda^3 - 5\lambda^2 + 8\lambda - 4.$$

By Theorem 1.26, the possible rational roots are $\pm 1, \pm 2$, and ± 4. We can test these by synthetic division to check if any is a root of the characteristic polynomial. Doing this shows that 1 is a root, so we have

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 1)(\lambda^2 - 4\lambda + 4) = (\lambda - 1)(\lambda - 2)^2 = 0$$

and therefore the distinct eigenvalues of A are $\lambda = 1, 2$.

[1] We write $a \mid b$ if a divides b; that is, if there exists a positive integer k such that $b = ka$.

Exercise 1.29. Show that the eigenspaces corresponding to the eigenvalues found in the previous example are

$$W_1 = \text{span}\left\{\begin{bmatrix}1\\0\\2\end{bmatrix}\right\},\ W_2 = \text{span}\left\{\begin{bmatrix}1\\1\\2\end{bmatrix}\right\}.$$